



Semester 2 Examinations 2021-22

Course Instance Code(s) 4BS2, 4BMS2, 4BME1, 4BCT1

Examinations Fourth Science

Module Codes MA4344

Module Advanced Group Theory

External Examiner(s) Prof C. Roney-Dougal

Internal Examiner(s) Dr. Niall Madden

Dr James Cruickshank

Duration 2 hours

No. of Pages 2 pages (including this cover page)

Discipline Mathematics

Instructions **Attempt ALL FOUR questions.** Each question carries 25 marks in total. Marks for individual parts of questions are indicated inline.

Requirements

Release in Exam Venue	Yes
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- Q1.** (a) What does it mean to say that G is the *internal direct product* of subgroups H and K [5 marks]?
 (b) Prove that $\mathbb{Z}_6$ is an internal direct product of subgroups H and K where $|H| = 3$ and $|K| = 2$ [10 marks].
 (c) Suppose that G is a group of order 21 and G has normal subgroups H and K such that $|H| = 3$ and $|K| = 7$. Prove that G is a cyclic group [10 marks].
- Q2.** (a) Suppose that G is a group, H is a subgroup of G and N is a normal subgroup of G . Prove that HN is a subgroup of G [6 marks]. State the Second Isomorphism Theorem [3 marks].
 (b) What is the definition of an *inner automorphism* of a group [3 marks]? Give an example, with proof, of a group G and an automorphism of G that is not an inner automorphism [6 marks].
 (c) Recall that $E(2) = \{(A, x) : A \in O(2) \text{ and } x \in \mathbb{R}^2\}$ with group operation $(A, x)(B, y) = (AB, x + Ay)$. Compute the inverse of the element
- $$\left(\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right)$$
- in the group $E(2)$ [7 marks].
- Q3.** (a) Draw (a representative portion of) a wallpaper pattern whose symmetry group contains a rotation of order 6 but does not contain any reflection [7 marks].
 (b) State the Fundamental Theorem of Finite Abelian Groups [3 marks] and use it to determine the number of distinct (up to isomorphism) abelian groups of order 200 [6 marks].
 (c) Suppose that a group G acts on a set X . What is the definition of the *orbit* of an element $x \in X$ [3 marks]? Find the conjugacy class of the permutation $(1, 2, 3)$ in the group S_4 [6 marks].
- Q4.** (a) State [3 marks] and prove [6 marks] Burnside's Counting Theorem.
 (b) Use Burnside's Counting Theorem to determine the number of distinct ways to colour the vertices of a regular pentagon with two colours. Here we say that two colourings are distinct if it is not possible to obtain one from the other by a symmetry of the pentagon [10 marks].
 (c) Prove that every group of order 45 has a normal subgroup of order 9 [6 marks].